

Module Interface Specification for Software Engineering

Team 11, technically functional

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1 Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

2 Symbols, Abbreviations and Acronyms

See SRS Documentation at [\[give url —SS\]](#)

[\[Also add any additional symbols, abbreviations or acronyms —SS\]](#)

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3 Introduction

The following document details the Module Interface Specifications for a Physiotherapy Companion app. The application is designed to identify users using computer vision for their prescribed physiotherapy exercise. In addition, an analysis will be performed on the user's movement and, after completion, the app will provide metrics to the user on their performance.

This tool utilizes OpenCV and MediaPipe for image recognition and then performs measurements on the number of repetitions, range of motion and perceived exertion.

Complementary documents include the [System Requirement Specifications](#) and [Module Guide](#). The full documentation and implementation can be found at <https://github.com/M9Huynh/technically-functional>.

4 Notation

[You should describe your notation. You can use what is below as a starting point. —SS]

The structure of the MIS for modules comes from [Hoffman and Strooper \(1995\)](#), with the addition that template modules have been adapted from [Ghezzi et al. \(2003\)](#). The mathematical notation comes from Chapter 3 of [Hoffman and Strooper \(1995\)](#). For instance, the symbol $:=$ is used for a multiple assignment statement and conditional rules follow the form $(c_1 \Rightarrow r_1 | c_2 \Rightarrow r_2 | \dots | c_n \Rightarrow r_n)$.

The following table summarizes the primitive data types used by Software Engineering.

Data Type	Notation	Description
character	char	a single symbol or digit
integer	$\mathbb{Z}$	a number without a fractional component in $(-\infty, \infty)$
natural number	$\mathbb{N}$	a number without a fractional component in $[1, \infty)$
real	$\mathbb{R}$	any number in $(-\infty, \infty)$

The specification of Software Engineering uses some derived data types: sequences, strings, and tuples. Sequences are lists filled with elements of the same data type. Strings are sequences of characters. Tuples contain a list of values, potentially of different types. In addition, Software Engineering uses functions, which are defined by the data types of their inputs and outputs. Local functions are described by giving their type signature followed by their specification.

5 Module Decomposition

The following table is taken directly from the Module Guide document for this project.

Level 1	Level 2
Hardware-Hiding	
Behaviour-Hiding	Input Parameters Output Format Output Verification Temperature ODEs Energy Equations Control Module Specification Parameters Module
Software Decision	Sequence Data Structure ODE Solver Plotting

Table 1: Module Hierarchy

6 Variables

Name	Type	Description
Name	String	This is the name of the user.
role	String	Values can be PT or patient
email	String	This will be used as their login ID
password	String	This is set by the user for login purposes
birthday	String	This is the user's birthday that will be used for verification purposes
license_no	Int	This is used for verification of PT
invite_code	String	This is given by the PT to the patient to allow registration on the application

Table 2: User Account Data Fields

7 MIS

7.1 Module - Account Management

ADT

This module's purpose is to initialize an instance of UserAccount and to handle any account-related tasks (e.g. Account creation (Patient), Account deletion (PT)).

7.1.1 Uses

- name
- role
- email
- password
- birthday
- license_no
- invite_code

7.1.2 Syntax

Exported Constants

- *UserAccount_Patient*: ADT
- *UserAccount_PT*: ADT

Exported Access Programs

Access Routine Name	Input	Output	Exceptions
account_create	<i>Name</i> <i>role</i> <i>birthday</i> <i>license_no</i> <i>invite_code</i> <i>email</i> <i>password</i>	UserAccount_Patient (ADT) or UserAccount_PT (ADT)	Input_format_invalid required_field_empty
account_delete	<i>Name</i> <i>email</i>	NA	input_format_invalid required_field_empty patient_not_found

Table 3: Account Management Access Routines

7.1.3 Semantics

State Variables

A: an instance of UserAccount

B: an instance of UserAccount

State Invariant

$\forall A, B \in UserDatabase : A[i] \neq B[j]$

$0 < A < length(A) \cap A[i] \notin \emptyset$

7.1.4 Assumptions

7.1.5 Access Routine Semantics

[accessProg —SS]():

- transition: [if appropriate —SS]

- output: [if appropriate —SS]
- exception: [if appropriate —SS]

[A module without environment variables or state variables is unlikely to have a state transition. In this case a state transition can only occur if the module is changing the state of another module. —SS]

[Modules rarely have both a transition and an output. In most cases you will have one or the other. —SS]

7.1.6 Local Functions

[As appropriate —SS] [These functions are for the purpose of specification. They are not necessarily something that is going to be implemented explicitly. Even if they are implemented, they are not exported; they only have local scope. —SS]

References

- Carlo Ghezzi, Mehdi Jazayeri, and Dino Mandrioli. *Fundamentals of Software Engineering*. Prentice Hall, Upper Saddle River, NJ, USA, 2nd edition, 2003.
- Daniel M. Hoffman and Paul A. Strooper. *Software Design, Automated Testing, and Maintenance: A Practical Approach*. International Thomson Computer Press, New York, NY, USA, 1995. URL <http://citeseer.ist.psu.edu/428727.html>.

8 Appendix

[Extra information if required —SS]

Appendix — Reflection

[Not required for CAS 741 projects —SS]

The information in this section will be used to evaluate the team members on the graduate attribute of Problem Analysis and Design.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing “what you think the evaluator wants to hear.”

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your design decisions stemmed from speaking to your client(s) or a proxy (e.g. your peers, stakeholders, potential users)? For those that were not, why, and where did they come from?
4. While creating the design doc, what parts of your other documents (e.g. requirements, hazard analysis, etc), if any, needed to be changed, and why?
5. What are the limitations of your solution? Put another way, given unlimited resources, what could you do to make the project better? (LO_ProbSolutions)
6. Give a brief overview of other design solutions you considered. What are the benefits and tradeoffs of those other designs compared with the chosen design? From all the potential options, why did you select the documented design? (LO_Explores)